

Mathematics — School Test Marking Report

Student: Seemab Atif (Roll 24683, M II-H)
School: OPF Girls College F-8/2, Islamabad
Paper: SSC-I (9th Grade) Mathematics, Chapter 4 — Factorization and Algebraic Manipulation
Format: BISE Annual Examination 2026 style, Total 65 marks, 2h 30m
School result on report card: 13/25 (52%, Grade D)
Date filed: 2026-05-25

Note: this is a diagnostic marking of the work visible in Seemab's answer sheet. Section A (MCQs) was not in the scanned pages, so MCQ marks are not assessed. The 13/25 grade on the school card may be a scaled version of a slightly different test; what follows is an honest read of the maths in the answer sheet itself.

Section B — Short Questions (11 × 3 = 33 marks)

Question	Her work	Diagnostic	Marks
Q.2(i) Factorize $2x^2y^3 - 6xy^2 + 2xy^3$	$= 2xy^2(xy - 3x + y)$	Common-factor extraction structurally right but the inside bracket doesn't balance after distribution. Sign/term mix-up. Partial.	1/3
Q.2(ii) OR Factorize $1 - x^2 - 2xy - y^2$	$= -1(x^2 - 2xy - y^2) = -1(x^2 - y^2)$	Wrong approach. Correct method: $1 - (x^2 + 2xy + y^2) = 1 - (x+y)^2 = (1-x-y)(1+x+y)$. She lost the $+2xy$ sign when factoring -1 .	0/3
Q.2(iii) Factorize $x^4 + 4m^4$	$(x^2)^2 + (2m^2)^2 \dots (x^2 + 2m^2)^2$	Sophie Germain identity started but not finished. Needs add & subtract $4x^2m^2$: $x^4 + 4x^2m^2 + 4m^4 - 4x^2m^2 = (x^2 + 2m^2)^2 - (2xm)^2 = (x^2 + 2m^2 - 2xm)(x^2 + 2m^2 + 2xm)$.	1/3
Q.2(iv) Factorize $y^2 - 7y + 12$	$y^2 - 3y - 4y + 12 = y(y-3) - 4(y-3) = (y-4)(y-3)$	Perfect. Clean middle-term break, correct factoring by grouping.	3/3
Q.2(v) Factorize $x^3 + 27y^3$	$(x+3y)(x^2 - 6xy + 9y^2)$	Wrong formula applied. Sum of cubes is $a^3+b^3 = (a+b)(a^2 - ab + b^2)$, so the middle term should be $-3xy$, not $-6xy$. She used $(a-b)^2$ expansion by mistake. Correct: $(x + 3y)(x^2 - 3xy + 9y^2)$	1/3
Q.2(vi) Factorize $8x^3 - 60x^2 + 150x - 125$	Mixed up grouping into $4x^2(2x-15) - 25(5-6x)$, then restarted as $2x^2(4x^2-30) - 5(25-30x)$	Did not recognise the cube pattern. This is $(2x - 5)^3 = 8x^3 - 3(2x)^2(5) + 3(2x)(5^2) - 5^3 = 8x^3 - 60x^2 + 150x - 125$. Correct: $(2x - 5)^3$	0/3
Q.2(vii) HCF of $(x+y)^2$ and $x^2 - y^2$	(1) $(x+y)^2 = (x+y)(x+y)$ ✓ (2) $x^2 - y^2 = (x+y)(x-y)$ ✓ HCF = $x+y$	Right answer. Initial expansion of $(x+y)^2$ had stray terms but she recovered with the correct factored form. Final HCF correct.	3/3
Q.2(viii) OR HCF of $x^2 - 49$ and $x^2 - 4x - 21$	(1) $x^2 - 49 = (x+7)(x-7)$ ✓ (2) $x^2 - 4x - 21 = (x+3)(x-7)$ ✓ HCF = $x - 7$	Fully correct. Difference of squares + middle-term break, both clean.	3/3
Q.2(ix) OR LCM of x^2+x-6 , x^2+x-2 , x^2-4x+3	Wrote $x \times x + x + 2 \times 3$ etc. Never factorised.	Did not attempt factoring. Should be: $(x+3)(x-2)$, $(x+2)(x-1)$, $(x-3)(x-1)$. LCM: $(x+3)(x-2)(x+2)(x-1)(x-3)$	0/3
Q.2(x) OR $(r^2 - s^2)/r \div (r-s)/s$	$\dots = (r + s^2)/r$	Wrong final form. Cancelled $(r-s)$ correctly but multiplied wrong. Correct: $s(r+s)/r$	1/3
Q.2(xi)	$(2a)^2 + 2(2a)(3b) + (3b)^2 = (2a + 3b)^2$	Perfect. Recognised perfect square	3/3

Square root of $4a^2 + 12ab + 9b^2$	$\sqrt{} = \pm(2a + 3b)$	trinomial cleanly.	
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Section B total: ~16 / 33

Section C — Long Questions (4 × 5 = 20 marks)

Question	Her work	Diagnostic	Marks
Q.3(i) Factorize $(x^2+3x-4)(x^2+3x+5) + 8$	Let $a = x^2+3x$. $(a-4)(a+5) + 8 = a^2 + a - 12 = (a+4)(a-3)$ Substitute back: $(x^2 + 3x + 4)(x^2 + 3x - 3)$	Excellent. Substitution method applied cleanly, expansion correct, middle-term break correct, back-substitution correct.	5/5
Q.3(ii) HCF of $P = 2m^3 - 2m^2n$, $Q = 4m^4 - 4m^2n^2$, $R = 2m^3 - 4m^2n + 2mn^2$	$P = 2m^2(m-n)$ $Q = 4m^2(m-n)(m+n)$ $R = 2m(m-n)^2$ HCF = $2m(m-n)$	Fully correct. All three factorisations clean, HCF extracted correctly as the lowest power of each common factor.	5/5
Q.3(iii) LCM and HCF of $2x^2-12xy+16y^2$, $x^2-6xy+8y^2$, $3x^2-12y^2$ (and verify $\text{LCM} \times \text{HCF} = P \times Q$)	Factored to $2(x-2y)(x-4y)$, $(x-2y)(x-4y)$ [wrote $(x-8y)$ once], $3(x-2y)(x+2y)$. LCM = $6(x-4y)(x-2y)(x+2y)$ ✓ HCF written as $(x-4y)(x-2y)$ ✗	LCM correct but HCF wrong. Common factor across all three is only $(x-2y)$ — the third polynomial $3(x-2y)(x+2y)$ does not have $(x-4y)$. Verification step then got tangled. Correct HCF: $(x-2y)$	3/5
Q.3(iv) OR Simplify $-2x/(x+3) + 3/(3-x) - (8x-12)/(x^2-9)$	Rewrote $3/(3-x)$ as $-3/(x-3)$, found common denominator $(x+3)(x-3)$, combined numerators step-by-step: $-2x^2 + 6x - 3x - 9 - 8x + 12 = -2x^2 - 5x + 3$ $= -2x(x+3) + (x+3)$ over $(x+3)(x-3)$ $= (-2x+1)(x+3)/((x+3)(x-3))$ $= (-2x+1)/(x-3)$	Excellent end-to-end. Sign handling, common denominator, expansion, factor recognition, and cancellation all clean. This is the hardest question on the paper and she nailed it.	5/5

Section C total: 18 / 20

Overall Score (Sections B + C only, MCQs not scanned)

Section	Score	Out of	%
B (Short)	~16	33	~48%
C (Long)	18	20	90%
B + C total	~34	53	~64%

If MCQs were ~50% (6/12), grand total $\approx 40/65 \approx 62\%$. The school's 13/25 (52%) likely reflects stricter marking or a different paper version — but the diagnostic shape is the same: **Section C strong, Section B is where the marks are being lost.**

What this tells us — and what to do next

The good news: Section C (the hardest, longest problems) is mostly solid. Q.3(i) substitution, Q.3(ii) multi-polynomial HCF, and Q.3(iv) algebraic-fraction simplification are all done correctly. This is high-skill work and shows real understanding of the chapter.

Where the marks went: Section B is the bleeding wound. Specific gaps:

- **Sum/difference of cubes formula confusion (Q.2(v))** — she used $(a-b)^2$ expansion instead of $a^3+b^3 = (a+b)(a^2 - ab + b^2)$. Single-formula fix.
- **$(2x - 5)^3$ not recognised as a cube (Q.2(vi))** — needs practice spotting the $(a-b)^3$ pattern: $a^3 - 3a^2b + 3ab^2 - b^3$.
- **Sign handling when factoring out -1 (Q.2(ii) OR)** — losing the $+2xy$ sign mid-rearrangement.
- **Sophie Germain identity unfinished (Q.2(iii))** — got the setup but didn't add & subtract the cross term.
- **LCM of three quadratics not attempted (Q.2(ix) OR)** — didn't try middle-term break on the simpler

ones, possibly time pressure.

- **Algebraic-fraction division final-step error (Q.2(x) OR)** — careless multiplication after cancellation.

Revision priority for next exam:

1. Drill the standard identities side-by-side: a^2-b^2 , $(a\pm b)^2$, $a^3\pm b^3$, $(a\pm b)^3$ — until she can produce them from memory without confusion.
2. Practice 10 mixed factorisations a day where she has to decide *which* identity applies before solving. The confusion isn't algebra ability, it's identity selection.
3. Sign discipline: 5 minutes a day on -1 factoring practice, e.g. rewrite $(1 - x^2 - 2xy - y^2)$ cleanly.
4. Time management: Section C took up bandwidth she needed for Section B. For 65-mark / 2h 30m papers, spend ~ 1 h on Section C, ~ 1 h 15m on Section B, and leave 15 min for MCQs.

Note for context: our home exam-04-ch4 covers the same chapter. Doing that paper as a follow-up under timed conditions (2h) would give us a clean apples-to-apples comparison after a week of identity drill.